

Amber (Jiale) Wang

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TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, Go, C/C++, SQL, PHP, Kotlin, MATLAB, R

Frameworks: React, React Native, Node.js, Express.js, Tailwind, TensorFlow, LangChain, Scikit-learn

Tools and Platforms: Unix, Azure, Selenium, MySQL, MongoDB, Mongoose, Git, Postman, Power Automate

EDUCATION

University of Waterloo

Master of Mathematics in Computer Science; Advisor: Pengyu Nies; GPA: 97/100

Waterloo, Ontario

Sept 2024 – Dec 2026

- Research Interests: Leveraging machine learning to improve software engineering, focusing on testing automation, code analysis using natural language processing, and data-driven development optimization.

Queen's University

Bachelor of Computing, Mathematics and Analytics; GPA: 4.0/4.3

Kingston, Ontario

Sept 2019 – Apr 2024

- Relevant Coursework: Algorithms, Data Structures, Software Architecture, Database Management, Computer Architecture, Operating Systems, Formal Methods in Software Eng, Statistics and Probability.

PUBLICATIONS

- **Jiale Amber Wang**, Kaiyuan Wang, and Pengyu Nie. Efficient Incremental Code Coverage Analysis for Regression Test Suites. In International Conference on Automated Software Engineering (ASE). 2024.
- Jerin Yasmin, **Jiale Amber Wang**, Yuan Tian, Bram Adams. An Empirical Study of Developers' Challenges in Implementing Workflows as Code: A Case Study on Apache Airflow. Journal of Systems and Software. 2024.

WORK EXPERIENCE

Machine Learning Developer

Royal Bank of Canada

Toronto, ON

May 2025 - Current

- Built and iterated on Generative AI prototypes using **Python**, **JavaScript**, and vector databases.
- Researched and tested **LLM** tools to recommend solutions for internal use cases.
- Contributed to Agile sprints by coding, testing, and deploying **ML workflows** and features.

Research Assistant

University of Waterloo

Waterloo, ON

Jan 2024 - Current

- Participated in writing a paper titled “Maintaining test cases using large language models”, demonstrating the potential of **large language models** to revolutionize test case maintenance in Java.
- Developed Python scripts to efficiently collect and preprocess code-test updates from GitHub repositories, showcasing strong **data engineering** skills.
- Implemented advanced **machine learning techniques**, including fine-tuning and multi-round refinement, to create a practical solution for automated test case updates, contributing to the software development community.

Automation Engineer

Ericsson Canada Inc.

Ottawa, ON

Sept 2022 - Dec 2023

- Created, developed, and maintained 10+ automated workflows using **Power Automate**, **Azure**, and **Python**. Generated 20+ error-free dashboards daily across departments, resulting in a **90% reduction** in manual processes. Conducted testing, debugging, and troubleshooting to ensure the consistent and accurate performance of all tools.
- Developed and implemented an automated email server that streamlined the license request process for an internal software tool, serving 400+ users.
- Developed an innovative web app using **JavaScript**, **CSS**, and **React.js**, featuring a cause-and-effect chain builder. Empowered users to visualize key drivers and outcomes in a chain diagram for problem-solving purposes.

Teaching Assistant

Queen's University

Kingston, ON

Sept 2021 - Aug 2023

- Instructed 200+ students on Data Structure, Introduction to Computer Science II, and Discrete Mathematics for Computing I; Covering data structures, algorithms, object-oriented software development, and proof methods.

- Assessed student performance in assignments, quizzes, and exams, successfully addressing 200+ coding errors in **Python** and **Java** programs and providing feedback for improvement.
- Responded to inquiry emails and course discussion forums frequently, improving students' conceptual understanding of course materials.

Advertising Data Analyst

Xiaomi Technology Co.

Beijing, China

May 2021 – Aug 2021

- Created and managed ad campaigns for Chinese overseas customers, offline customers, and agencies; Regularly promoted and monitored advertisements; Output daily/monthly reports and seasonal reviews.
- Assembled and analyzed promotion data with **SQL**; Optimized promotion by setting tiered pricing, customizing targets, etc., conversion of optimized campaigns increased **2-3** times.
- Responded to customers' inquiries, and provided promotion advice, the favorable rate is **100%**.

PROJECT EXPERIENCE

Software Developer, InQubate

Sept 2023 - Apr 2024

- Developed an **AI**-based language learning mobile application using **Type Script** and **React Native** that aims to improve users' pronunciation correctness and speech confidence.
- Integrated **OpenAI API** and **Azure Speech Services API** to create a voice-enabled chatbot and provide real-time feedback on user's pronunciation and sentence structure.
- Collaborated with business analysts and conducted pitches at various venture events to showcase the application's capabilities and potential impact, demonstrating strong collaboration and communication skills.

Tour Booking Web App

May 2023 - Aug 2023

- Developed a fully functional tours booking web application using **Node.js**, **Express.js**, **MongoDB**, and **Mongoose** for back-end, **HTML**, **CSS**, and **Pug** for front-end.
- Ensured top-tier security with bcrypt password encryption and **JWT**-based user authentication, providing different database management permissions based on user roles.
- Handled seamlessly payments through **Stripe.js** library and extended user sessions via secure cookie transmission.
- Deployed Node.js applications to production using **Heroku**, ensuring scalability and availability.

Borealis AI Undergraduate Mentorship Program

Oct 2021 - Jan 2022

- Used a dataset containing 500 redditors' posts with 5-label depression classification from subreddit r/depression to forecast depression level and risk of suicide through analyzing texts from a person.
- Employed **natural language processing** techniques, such as tokenization, stop word removal, lemmatization, and stemming, along with **Python** libraries like **NLTK** and **Scikit-learn**.
- Built and trained four **machine learning models**, outperformed existing published works in depression classification.

WeChat Mini-Program

Sept 2020 - Jan 2021

- Developed a WeChat mini program platform that provides learning and life information for college students in North America; Successfully gained over **200 active users** since its release.
- Implemented features such as post categorization, menu bar, and search functionality using **JavaScript**, **CSS**, and **HTML**; Utilized **MongoDB** as the database for storing and managing application data.